

● Preference seeds from natural-language input

○ Posterior bundle samples

● Selected bundle

□ Top-value bundles

Stage 1: Preference Elicitation

Clean description

Paraphrased description

Low ambiguity,
Rev. 412.7

Stage 2: Auction Decision

(a) Bidder Case 1

(b) Bidder Case 2

(c) Bidder Case 3

(d) Bidder Case 4

Selected bundle {1,3,5}
Price 72.4, Utility 18.1

Selected bundle {2,4}
Price 66.2, Utility 14.6

Selected bundle {1,2,3,5}
Price 88.3, Utility 22.4

Selected bundle {2,5,6}
Price 92.0, Utility 19.3

High ambiguity,
Rev. 381.4

Selected bundle {1,3,5}
Price 72.4, Utility 16.3

Selected bundle {2,4}
Price 66.2, Utility 12.7

Selected bundle {1,2,3,5}
Price 88.3, Utility 19.1

Selected bundle {2,5,6}
Price 92.0, Utility 16.4

Entropy 0.12

Entropy 0.24

Ambiguous description

Recovered posterior

Entropy 0.41

Entropy 0.19